PS1

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1. Proof

   - Associativity $(f*g)*h = f*(g*h)$

     Proof. By the definition of convolution,

     $$(f*g)(x) = \int_{-\infty}^{\infty} f(t)g(x-t)\,dt$$

     Now consider the convolution of $(f*g)$ with h:

     $$\begin{aligned}((f*g)*h)(x) &= \int_{-\infty}^{\infty} (f*g)(t)h(x-t)\,dt \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(s)g(t-s)\,ds\right] h(x-t)\,dt\end{aligned}$$

     Changing the order of integration gives

     $$\int_{-\infty}^{\infty} f(s)\left[\int_{-\infty}^{\infty} g(t-s)h(x-t)\,dt\right] ds$$

     Let $r = t-s$, so $t = r+s$ and $dt = dr$. Then

     $$\begin{aligned}((f*g)*h)(x) &= \int_{-\infty}^{\infty} f(s)\left[\int_{-\infty}^{\infty} g(r)h(x-s-r)\,dr\right] ds \\ &= \int_{-\infty}^{\infty} f(s)(g*h)(x-s)\,ds \\ &= (f*(g*h))(x)\end{aligned}$$

     Therefore,

     $$\boxed{(f*g)*h = f*(g*h)}.$$

     □

   - Distributivity $f*(g+h) = f*g + f*h$

     Proof. By the definition of convolution,

     $$\begin{aligned}f*(g+h))(x) &= \int_{-\infty}^{\infty} f(t)(g+h)(x-t)\,dt \\ &= \int_{-\infty}^{\infty} f(t)\left[g(x-t)+h(x-t)\right] dt\end{aligned}$$

     Using the distributive property,

$$f*(g+h))(x) = \int_{-\infty}^{\infty} f(t)g(x-t)\,dt + \int_{-\infty}^{\infty} f(t)h(x-t)\,dt$$
$$= (f*g)(x) + (f*h)(x)$$

Therefore,

$$\boxed{f*(g+h) = f*g + f*h}$$

□

- Differentiation rule

Proof. The differentiation rule for convolution is

$$\frac{d}{dx}(f*g) = f*g'$$

Starting from the definition of convolution,

$$f*g)(x) = \int_{-\infty}^{\infty} f(t)g(x-t)\,dt.$$

Differentiate both sides with respect to x:

$$\frac{d}{dx}(f*g)(x) = \frac{d}{dx}\int_{-\infty}^{\infty} f(t)g(x-t)\,dt$$
$$= \int_{-\infty}^{\infty} f(t)\frac{d}{dx}g(x-t)\,dt$$
$$= \int_{-\infty}^{\infty} f(t)g'(x-t)\,dt$$

By the definition of convolution,

$$\int_{-\infty}^{\infty} f(t)g'(x-t)\,dt = (f*g')(x)$$

Therefore,

$$\boxed{\frac{d}{dx}(f*g) = f*g'}$$

Similarly,

$$\boxed{\frac{d}{dx}(f*g) = f'*g}$$

□

- Convolution theorem

Proof. The convolution theorem states that

$$\mathscr{F}\{f*g\} = \mathscr{F}\{f\}\mathscr{F}\{g\}.$$

Let

$$F(\omega) = \mathscr{F}\{f\}, \qquad G(\omega) = \mathscr{F}\{g\}.$$

By the definition of the Fourier transform,

$$\begin{aligned}\mathscr{F}\{f*g\}(\omega) &= \int_{-\infty}^{\infty} (f*g)(x) e^{-i\omega x}\, dx \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) g(x-t)\, dt \right] e^{-i\omega x}\, dx.\end{aligned}$$

Changing the order of integration gives

$$= \int_{-\infty}^{\infty} f(t) \left[\int_{-\infty}^{\infty} g(x-t) e^{-i\omega x}\, dx \right] dt.$$

Let $y = x - t$. Then $x = y + t$ and $dx = dy$:

$$\begin{aligned}\mathscr{F}\{f*g\}(\omega) &= \int_{-\infty}^{\infty} f(t) \left[\int_{-\infty}^{\infty} g(y) e^{-i\omega(y+t)}\, dy \right] dt \\ &= \left[\int_{-\infty}^{\infty} f(t) e^{-i\omega t}\, dt \right] \left[\int_{-\infty}^{\infty} g(y) e^{-i\omega y}\, dy \right].\end{aligned}$$

The first integral is $F(\omega)$ and the second is $G(\omega)$. Hence,

$$\boxed{\mathscr{F}\{f*g\}(\omega) = F(\omega)G(\omega)}$$

Therefore, convolution in the spatial domain corresponds to multiplication in the frequency domain.

□

2. Frequency smoothing



(a) 7^2 frequencies



(b) 15^2 frequencies



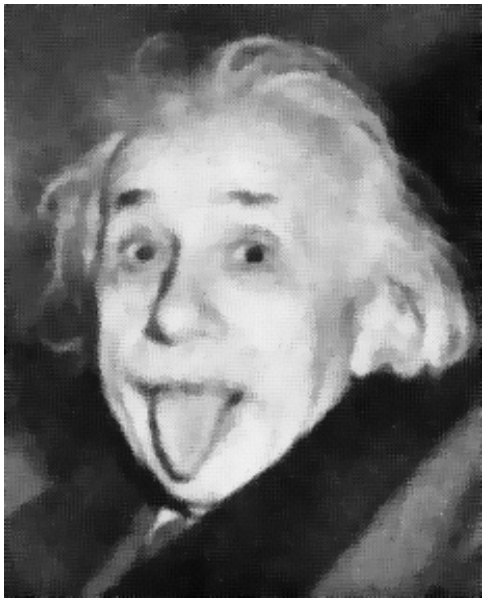
(c) 31^2 frequencies



(d) Full dimension

Figure 1: Frequency smoothing results

3. ROF



(a) 0.001 noise $\alpha = 0.008$ $\lambda = 4$

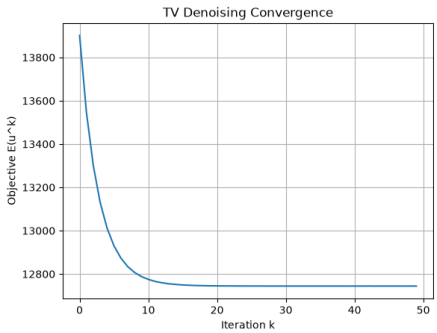


(b) 0.05 noise $\alpha = 0.008$ $\lambda = 0.8$



(c) 0.1 noise $\alpha = 0.008$ $\lambda = 0.5$

Figure 2: Result



(a)

Figure 3: One of the convergence Graphs